Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Precise Laser Slitting System for Secondary Battery Anode*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (X), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical and Electronics		*Electrical and Electronics*	
	Medium Category	Small Category	Medium Category	Small Category
	Optical Application Devices	*Laser Processing Machines*	*Batteries*	*Manufacturing and Measurement/Evaluation Equipment*
Concept of Proposed Technology	*The existing slitting process uses mechanical circular knives (slitting knives) for cutting.* *This method has drawbacks: the quality is inconsistent depending on the initial and later stages of knife use, and the resulting burrs and particles cause electrode damage and cell performance degradation.* *Pre-studies were performed and a conference paper was published in Korea.* *The results showed that while conventional mechanical knives produced slitting burrs of 21.5 μm at a production speed of 60 m/min and 43.2 μm at 120 m/min, the use of a laser slitter significantly reduced the size of the slitting burrs to 6.7 μm at 60 m/min and 8.7 μm at 120 m/min.* *This study confirmed the feasibility of using lasers for slitting secondary battery anodes and is expected to contribute to the high-precision and high-reliability of future electrode manufacturing processes.*			

Rationale for Joint R&D Project	*Due to the increase in demand for secondary batteries, battery manufacturers worldwide are pursuing technology development and process innovation to produce high-efficiency, high-quality batteries.* *To achieve such process innovation, the slitting process, which affects the dimensional accuracy and surface quality of the electrodes, is a core process that directly impacts battery performance and efficiency. Also, a feasible laser process for a stable production process must be developed first.* *The implementation of laser slitting for mass production requires the integration of laser source technology, optical system optimization, and roll-to-roll manufacturing systems, which is difficult to achieve within a single organization alone.* *Therefore, an international joint R&D project is required to combine complementary expertise and accelerate the development of an industrially viable laser slitting solution.*
Key Words	*1. Laser Slitting* *2. Secondary Battery Electrode Manufacturing* *3. Low-Burr Cutting* *4. Heat Affected Zone (HAZ) Minimization* *5. Roll-to-Roll Processing*

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (4)	**End of the Project**	Level (7-8)
R&D Objectives	*The objective of this R&D project is to develop and industrialize a high-precision laser slitting system for secondary battery anode electrodes that can replace conventional mechanical slitting processes.* *The specific objectives are as follows:* *1. To develop a laser slitting process capable of significantly reducing slitting burrs and minimizing the heat affected zone (HAZ)* *2. To optimize laser source and optical system configurations suitable for high-speed electrode slitting* *3. To design and integrate a roll-to-roll laser slitting system applicable to mass production environments*			
R&D Target Specifications	*1. Reduction of slitting burr size by at least 80% compared to conventional mechanical slitting* *2. Minimization of heat affected zone (HAZ) to a level that does not cause material degradation* *3. Stable laser slitting operation at production speeds of 60–120 m/min or higher* *4. Consistent cutting quality independent of tool wear*			
R&D Plan	*1. Development of laser slitting machine and real production* *2. Development of fast and precise laser cutting (slitting) roll system* *3. Development of laser optics and mechanical roll system*			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*Contribution to the development of laser slitting process principles and core laser processing technologies based on expertise in laser–material interaction.* *Support for the optimization of laser parameters and optical system configurations to reduce slitting burrs and minimize the heat affected zone (HAZ).* *Investigation of approaches to scale up the productivity to higher cutting speeds as 120m/min via multi beam optical systems or beam shaping approaches (optical stamping of the slits)* *Technical support for performance analysis and evaluation of laser slitting results.* *Knowledge sharing and technical consultation within the framework of international joint R&D collaboration.*
Expected R&R of Korean SME Partner	*Development of a laser slitting machine and roll-to-roll mechanical system applicable to secondary battery electrode production.* *Integration of laser, optical, and mechanical systems into a practical laser slitting system.* *Execution of pilot-scale testing and verification under production-relevant conditions.* *Contribution to system validation, manufacturability assessment, and preparation for industrial application.* *Active participation in international collaboration and coordination with the research institute.*

5. Expected Outcomes/Impact

Expected Outcomes and Impact	*This joint R&D project is expected to achieve a high-precision laser slitting system for secondary battery anode electrodes with significantly improved cutting quality compared to conventional mechanical slitting processes.* *From a technological perspective, the project will enable:* *- Significant reduction of slitting burrs and minimization of the heat affected zone (HAZ), leading to improved electrode edge quality.* *- Stable and consistent slitting performance independent of mechanical tool wear.* *- Validation of laser-based slitting technology for secondary battery electrode manufacturing under production-relevant conditions.* *From an industrial and economic perspective, the developed technology is expected to:* *- Contribute to the production of high-quality and reliable secondary batteries by improving electrode manufacturing processes.* *- Enhance manufacturing competitiveness of Korean SMEs through the adoption of advanced laser processing technologies.* *- Support the development of safer batteries for EV and ESS applications by improving electrode quality and process reliability.*

6. References

(1) Pre-study results on laser slitting technology for secondary battery electrodes.

(2)

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization